When the Public Leaderboard Misleads: Lessons from 99 Scored Submissions in Single-Subject MEG Word Decoding

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Abstract

We describe our entry to the Deep track of the 2026 PNPL Competition, which asks for predictions of which of 50 English words a participant hears, made from magnetoencephalography (MEG) recordings and scored by class-balanced top-10 accuracy (BAcc@10). Our system combines an ensemble of convolutional word encoders and sentence-level Transformers with a closed-vocabulary language-model (LM) prior. Kaggle's private leaderboard ranked each team by the best private score among ten submissions it selected. In the last week, an LM fine-tuned on synthetic vocabulary-only sentences and whole-sentence candidate decoding raised our public score to 0.733 (13th of 91 teams). On the private split, however, the best of our ten selected submissions scored 0.663 (44th), while an earlier and simpler submission had reached 0.688. Using the public and private scores of all 99 of our submissions, we examine this gap retrospectively. Four large early changes kept the sign of their effect from the public to the private split; only 10 of 26 later changes did. Among submissions with public score at least 0.66, those from the last week averaged 0.714 public and 0.648 private, against 0.686 and 0.666 for earlier ones, and within either group the two scores were uncorrelated. Three repeats of one recipe with different seeds differed by up to 0.015 on the private split, more than most effects we tested. Only 0.24% of random sets of ten drawn from the same 60 submissions did as badly as the ten with the highest public scores. These are exploratory observations from one development history, from which we draw practical recommendations for choosing final submissions when the public split is small.

1 Introduction

In the Deep track, all data come from one participant (`subj0`). Each test item is a word the participant heard, either within a sentence or in isolation. A system outputs a distribution over the 50 target words, and the score is the mean over the 50 words of each word's Recall@10, so chance is 0.20. The training data are about 780k words of audiobooks, TIMIT, MOCHA-TIMIT and podcasts. The test sentences are short conversational utterances made almost entirely of the 50 target words [2].

During the competition, Kaggle shows a public score computed on part of the test set. Each team then selects ten submissions, and Kaggle's private leaderboard ranks it by the highest private score among them. The organisers determine the final standings separately, on independent holdout data recorded with different stimuli; all ranks in this paper are Kaggle's. We relied heavily on the public score, submitting up to ten times a day, which may have exposed us to adaptive over-selection. Kaggle released the private scores of all our submissions, so we can study retrospectively how public-score gains and our final selection related to private performance.

2026 PNPL Competition.

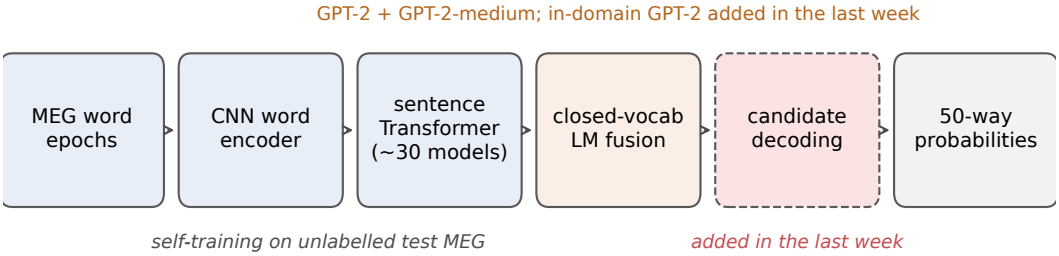


Figure 1: System overview. Word epochs pass through a per-word CNN and a sentence Transformer; the ensemble's 50-way log-probabilities are combined with a language-model prior and, in the last week, with whole-sentence candidate decoding (dashed).

The paper has three parts: the system (Section 2), a retrospective public–private analysis (Section 3), and a back-test of final-selection rules with recommendations (Section 4). All leaderboard statistics are computed from the Kaggle export of our submissions by a released script. Hand-coded inputs, namely which submissions are compared, their labels and the two selections, are listed in that script with their sources in our dated experiment log.

2 System

Data and validation. A run is one continuous recording, and a session is one recording visit. Each run is z-scored per channel and projected by a global, whitened PCA from 306 channels to 80 components. A word epoch is the window around a word's onset: $[-0.5, 2.5]$ s at 50 or 100 Hz ("3 s" models) or $[0, 1]$ s at 125 Hz ("1 s" models). TIMIT SA1/SA2 sentences recur 630 times each, and sentence models memorise them, so we train repeated sentences as isolated words.

Validation uses seven runs held out from supervised training (the PCA was fitted on a sample of runs that included two of them). We report two validation measures on these runs, both in BAcc@10. *Novel-sentence validation* uses held-out sentences whose text never occurs in training. The *proxy* uses 312 segments of these runs with 1,042 occurrences of the 50 target words, so that LM fusion can be tested. The test recordings were made in separate sessions, and their per-component input scale is about 1.35 times that of training.

MEG decoder. We follow the word-decoding setup of d'Ascoli et al. [1], without their semantic objective. A convolutional encoder maps each word epoch to an embedding, and a pre-norm Transformer over the words of a sentence classifies each word among 1,000 classes: the 50 target words and the 950 most frequent other words. The submitted probabilities are a softmax over the 50 target classes only. The base configuration has d=512 and 8 layers, and a second head classifies each word without context. Final ensembles average the log-probabilities of about 30 members, which differ in window, sampling rate, depth (4 to 16 layers), CNN width and input normalisation. IN1 denotes our 31-member ensemble that includes three instance-normalised models. Some members are self-trained on the unlabelled test recordings, using an earlier submission's predictions as soft targets.

Language prior. For each sentence position, a causal LM gives the probability of each of the 50 words, averaged over the beam hypotheses for the preceding words. Following pointwise mutual information (PMI), we subtract, for each word, the log of its average LM probability over all test positions. The result is added to the MEG log-probabilities with weight 0.5, using beam search with 32 hypotheses. GPT-2 [3] and GPT-2-medium are combined after rescaling their PMI terms to the same standard deviation.

Prior correction, word timing and V. *Prior correction* with strength τ subtracts τ times the log training frequency of each word from the MEG log-probabilities. *Word-timing evidence* with weight β adds β times the log-likelihood of the gap to the next word's onset given the word. The *validation-motivated stack V* combines input rescaling, word timing ($\beta = 0.5$) and sentence prior correction ($\tau = 0.3$).

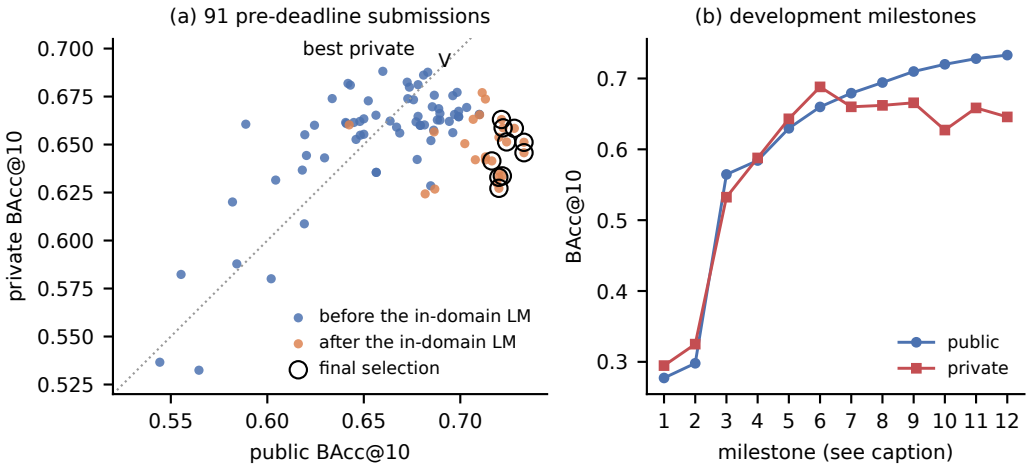


Figure 2: (a) Public and private BAcc@10 of the 91 pre-deadline submissions with public score at least 0.5 (dotted line: equal scores), coloured by whether they were made before or after the in-domain LM was introduced. Black rings mark our ten final selections. (b) Milestones: 1 linear baseline, 2 per-word CNN, 3 sentence Transformer, 4 repeated-sentence handling, 5 closed-vocabulary LM fusion, 6 the 9-model ensemble with our best private score, 7 3 s windows and LM calibration, 8 self-training, 9 diverse ensemble B5, 10 in-domain LM (807 sentences, the lowest of three seeds on private), 11 IN1 with the 3,742-sentence LM, 12 candidate decoding. Each milestone may change several components.

Components added in the last week. The *in-domain LM* is GPT-2 fine-tuned on corpora of 807 to 3,742 synthetic sentences written by Claude and Qwen3.5-9B [4], each using only the 50 target words. One variant also adds a fine-tuned Qwen2.5-1.5B. *Whole-sentence candidate decoding* scores every candidate sentence with the same number of words as the test sentence by its summed MEG log-probability. A softmax over the candidates gives per-position word probabilities, which are mixed into the submission with weight λ. If fewer than 20 candidates of that length exist, the base prediction is kept. Generic candidates come from the synthetic corpora ($\lambda = 0.15$). Our best public submission adds LLM proposals prompted with our six highest-scoring words per position, a prior of 0.5 times each candidate's in-domain LM log-probability, and $\lambda = 0.3$.

Isolated words. For the 92 isolated words we averaged the contextual and context-free heads and applied prior correction ($\tau = 0.7$). During the competition we assumed that these rows counted only towards the private score. However, four pairs of submissions differed only in these rows, and in each pair the public and the private scores were identical to five decimals, although in the three pairs whose files we kept, 33–82% of these rows had different top-10 sets. The rows therefore appear not to be scored in either split; at least, our tuning of them had no measurable effect.

3 Public versus private

Scores of all submissions. Over the 95 pre-deadline submissions, public and private scores correlate at $r = 0.92$, but this reflects the large early steps. The correlation is 0.50 once the four sub-0.5 baselines are excluded, 0.11 for the 85 submissions with public score at least 0.6, and -0.37 for the 60 at or above 0.66 (95% interval -0.57 to -0.13); we chose these thresholds after seeing the data. The negative value is a contrast between two groups (Figure 2a). Among the 60, the 24 submissions from the last week averaged 0.714 public and 0.648 private, against 0.686 and 0.666 for the 36 earlier ones, and within each group the correlation is close to zero (0.17 and 0.01). Our best private score, 0.6881, came from a 9-model ensemble submitted on 9 October with public score 0.65986, and no later submission exceeded it (Figure 2b). Four further submissions, ablations of candidate decoding made after the deadline, are excluded throughout.

Table 1: Selected reference–variant comparisons (change in BAcc@10, variant minus reference) and the direction suggested by in-session validation before the private release (+ helps, − hurts; n/a: no documented validation difference larger than the proxy noise of about 0.015). † marks bundles of several changes. Appendix A lists all 30.

Change (variant minus reference)	Δ public	Δ private	Validation
sentence-context model (redesign)†	$+0.267$	$+0.207$	n/a
closed-vocab LM fusion (PMI)	$+0.045$	$+0.055$	$+$
Qwen3-8B instead of GPT-2	-0.041	-0.079	$-$
N-best hard marginals, no PMI†	-0.074	-0.061	n/a
self-training round 1	$+0.015$	$+0.002$	n/a
LM fusion after self-training	$+0.022$	-0.012	n/a
all members self-trained (one teacher)†	-0.014	-0.049	n/a
sentence-end LM term	-0.051	$+0.019$	n/a
sentence prior correction τ=0.3	-0.037	$+0.017$	$+$
word-timing evidence β=0.5	-0.060	-0.010	$+$
validation-motivated stack V†	-0.027	$+0.022$	$+$
in-domain LM (807 sentences, seed 0)	$+0.010$	-0.038	n/a
in-domain LM, 3-seed mixture†	$+0.010$	-0.012	n/a
in-domain corpus 807 → 1,368	-0.010	$+0.038$	n/a
in-domain corpus 807 → 2,103 (adds Qwen; IN1)	$+0.002$	$+0.018$	n/a
generic candidates, λ=0.15	-0.003	$+0.012$	n/a
candidate decoding (guided pool + LM prior)†	$+0.009$	-0.006	n/a

Why later public gains did not transfer. Three mechanisms make public differences between related submissions unreliable here. First, almost every new submission was built from the one with the best public score so far: for 29 of our 30 comparisons below, the reference was the running public maximum. Its public score partly reflected favourable noise that its variants did not inherit, so their public scores tended to fall back while their private scores did not (mean change -0.0095 public against -0.0003 private over the 26 later comparisons). Second, self-training and guided candidates copy a public-selected submission's test predictions into later submissions, so the later submissions are not independent. Third, both splits are small samples. Three seeds of one recipe gave private scores of 0.627, 0.630 and 0.642 (standard deviation 0.008), so a difference between two submissions needs to exceed about 0.022 before its sign is informative. 23 of our 30 private effects are smaller than that. Public scores repeat the same five-decimal values more often than private ones (81 against 90 distinct values among 95 submissions); for example, three different in-domain-LM submissions all scored 0.72001. This is consistent with a small public split, although we do not know its size.

Paired comparisons. We specified 30 reference–variant comparisons between submissions: 15 change one documented component and 15 bundle several (Table 1, Figure 3a, Appendix A). The four early structural changes, made between 7 and 9 October to a much weaker system, kept their sign on the private split: the sentence-context model ($+0.207$ private), LM fusion ($+0.055$), and the two LM alternatives we rejected. Among the 26 later comparisons, only 10 kept their sign (Wilson 95% interval 0.22–0.57), and the public and private effects were not positively related ($r = -0.29$, interval -0.61 to 0.11). Even later changes with an absolute public effect of at least 0.03 split two and two. The comparisons share references and were chosen by us, so these counts describe our history rather than a calibrated threshold.

The language components. Replacing GPT-2 with the 807-sentence in-domain LM increased the public score by 0.010 and decreased the private score by 0.038 for one seed. Over the three seeds the private change was -0.033 on average, and a mixture of the three seeds lost 0.012. Larger corpora mostly recovered this loss but did not clearly exceed the reference without the in-domain LM. Adding the 807 Claude sentences to the 1,575 Qwen sentences even reduced the private score from 0.674 to 0.651 while raising the public score. Across in-domain-LM submissions the private score ranged from 0.624 to 0.677. The 807-sentence variants with public scores of 0.719–0.722

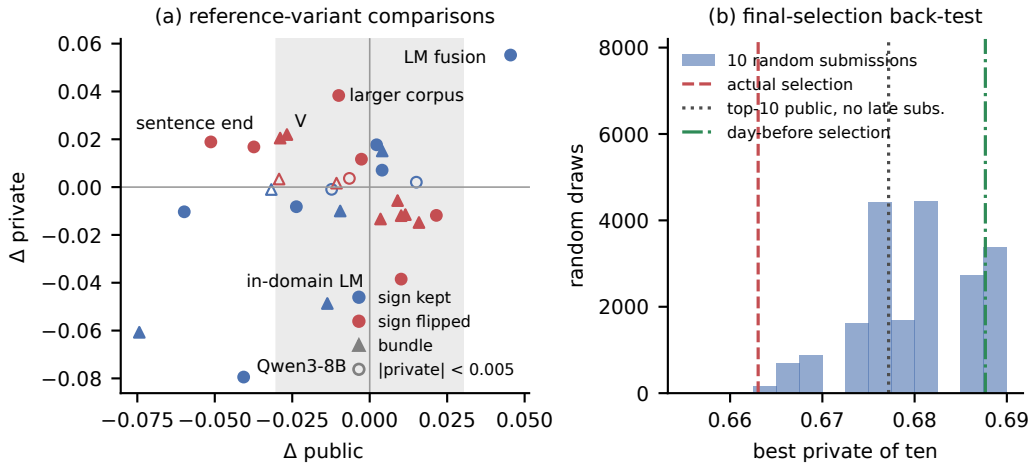


Figure 3: (a) Public and private effect of the 30 reference–variant comparisons; the sentence-context redesign ($+0.267/+0.207$) is off the scale. Grey band: absolute public effect below 0.03. Red: the sign changed between splits; triangles: bundles; open markers: private effect below 0.005. (b) Best private score of ten distinct submissions drawn at random (20,000 draws) from the 60 pre-deadline submissions with public score at least 0.66, with three selections from Table 2.

Table 2: Back-test of final-selection rules: best private score among ten submissions made before the deadline. Random rules: exact expectation and probability of scoring at most 0.663, over all sets of ten distinct submissions from the pool.

Rule	Best private	P(≤ 0.663)
Our actual final selection	0.663	–
Ten highest public scores (8 of 10 shared with the actual selection)	0.663	–
Ten highest public among submissions before the in-domain LM	0.677	–
Ten random from the 60 with public ≥ 0.66	0.680	0.24%
Ten random from the 24 last-week submissions among those 60	0.672	9.4%
Our selection one day before the deadline (included V)	0.688	–

scored 0.627–0.636 on private, among the lowest. Generic candidates changed the private score by $+0.012$, and the guided recipe with the best public score by -0.006, both within the seed spread. In short, the data neither establish nor rule out a benefit of these components, but they do show that selecting their variants by public score picked poor ones.

Validation against public. For seven comparisons, in-session validation and the public score pointed in opposite directions before the private release. The private split sided with validation three times ($\tau = 0.3$, the 1 s share and V) and with the public score four times ($\tau = 0.5$, the wide CNN and two word-timing strengths). Following validation would have changed the private score by $+0.003$ on average (95% interval -0.008 to $+0.014$; exact signed-rank $p = 0.81$), with every individual effect below the seed-noise level. The cases also overlap, since V contains $\tau = 0.3$ and word timing. Validation was therefore neither clearly better nor clearly worse than the public score. During the competition we read the repeated disagreements as evidence against the proxy and followed the public score.

4 Final selection

One day before the deadline, our recorded selection mixed four in-domain-LM submissions with three earlier ensembles and three validation-backed submissions, including V. On the last day, after the 3,742-sentence LM and candidate decoding raised the public score further, the final selection replaced all six submissions made before the in-domain LM with later ones. That moved the result

from 0.6877 to 0.6630. On Kaggle's private leaderboard, V's score would have placed us 34th instead of 44th.

Table 2 and Figure 3b compare simple alternatives. Choosing the ten highest public scores also gives 0.663. Only 0.24% of random sets of ten from the 60 submissions with public score at least 0.66 were as low; the figure is between 0.15% and 0.33% for thresholds from 0.64 to 0.68. Within the 24 last-week submissions, the family that the final selection came from, 9.4% were as low. V is the best private score in the 60-submission pool, and the overall best submission (public 0.65986) is just below the pool's cutoff. The pool and rules are defined in hindsight, so these figures describe our submissions rather than a general result.

Our recommendations for small public splits:

1. Repeat at least one recipe with different seeds, and treat public differences smaller than that spread, or than the spread among related configurations (here 0.01–0.03), as unreliable, especially for late changes chosen on the public split alone.
2. Diversify the final selection across selection criteria. When the result is the best of ten, spreading the selection over imperfectly correlated criteria (public score, validation, earlier strong systems) protects against any single criterion being wrong, as our day-before selection did.
3. Record each submission's reference submission, so that paired effects, and not only levels, can be checked afterwards.
4. Check which test rows are scored, for example by submitting pairs that differ only in some rows. Our isolated-word tuning had no measurable effect.

5 Discussion

For this single, deeply sampled subject, two large structural changes held up on the private split: the sentence-context redesign ($+0.207$) and closed-vocabulary LM fusion ($+0.055$). For most later changes, the direction of the public-score change did not predict the direction of the private-score change, and most private effects were below seed noise. The language-side components were neither shown to help nor ruled out, but choosing their variants by public score selected poor ones. A validation proxy built from recordings outside the training sessions would be a promising improvement.

Reproducibility. The repository `https://github.com/kuotunyu/pnpl2026-meg-word-decoding` contains the code, the corpora, the submission export with leaderboard snapshots, and the analysis script, and links to the checkpoints. MEG models were trained on Google Colab A100 GPUs and a local RTX 4090.

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References

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A All reference–variant comparisons

Table 3 lists all 30 comparisons with the Kaggle file names of the reference and the variant. Validation labels are component-level: they come from smaller validation ensembles, were recorded before the private release, and were sometimes recorded after the public result; the experiment-log entry behind each label is listed in the released analysis. Some references were submitted after their variant, as matched controls. With an absolute public-effect threshold of 0.025, 0.03 or 0.035, the large and small groups kept their sign in 6 of 11 and 8 of 19, 6 of 8 and 8 of 22, or 5 of 7 and 9 of 23 cases. Our two recorded selections are, on 15 October: `IN1_Q_llm_plus`, `B5i_L1_IN1`, `B5i_L1`, `B9c_L1`, `B5i`, `B9c`, `M1`, `B5d10`, `V`, `F1`; and on 16 October (final): `AVG_SCGL_Qbig`, `IN1Qp_SCGL_b0.5`, `IN1_Q_big`, `IN1_Q_llm_plus`, `B9cIN_Qplus`, `B5i_L1_IN1`, `IN1Qp_SC0.15`, `B5i_L1`, `B9c_L1`, `IN1_Q_big_SCG2L_l0.5` (file names without the `final_` prefix). The tables and statistics are generated by `analysis/analyze_lb.py` and `analysis/make_tables.py` from `analysis/data/submissions_raw.csv`.

Table 3: All 30 reference–variant comparisons (change in BAcc@10, variant minus reference). Type: single documented change or bundle of several. Validation as in Table 1.

#	Change	Reference	Variant	Type	Δ public	Δ private	Validation
1	sentence-context model (redesign)	`sub_v1`	`sub_c1`	bundle	$+0.267$	$+0.207$	n/a
2	closed-vocab LM fusion (PMI)	`sub_c5`	`sub_c5_lmpmi05`	single	$+0.045$	$+0.055$	$+$
3	N-best hard marginals, no PMI	`sub_c5_lmpmi05`	`sub_c5_lm05`	bundle	-0.074	-0.061	n/a
4	Qwen3-8B instead of GPT-2	`final_gpt2`	`final_qwen3_8b`	single	-0.041	-0.079	$-$
5	3 s window ensemble	`final_gpt2`	`finalW_gpt2`	bundle	$+0.016$	-0.015	n/a
6	GPT-2-medium added, strength-calibrated	`finalW_gpt2`	`finalW_calib2`	bundle	$+0.003$	-0.013	n/a
7	sentence prior correction τ=0.5	`finalW_calib2`	`finalW_calib2_t0.5`	single	-0.012	-0.001	$+$
8	higher 1 s share	`finalW_calib2_iso14`	`finalY_calib2_t0`	bundle	-0.029	$+0.003$	$+$
9	self-training round 1	`finalW_calib2_iso14`	`final_P1`	single	$+0.015$	$+0.002$	n/a
10	LM fusion after self-training	`final_P1noLM`	`final_P1`	single	$+0.022$	-0.012	n/a
11	self-training round 2 (all replaced)	`final_P1`	`final_P2`	bundle	-0.010	-0.010	n/a
12	sentence-end LM term	`final_P1`	`final_P1eos`	single	-0.051	$+0.019$	n/a
13	larger diverse ensemble	`final_P1`	`final_B2`	bundle	$+0.004$	$+0.015$	n/a
14	12-layer + wide-CNN members	`final_B2`	`final_B5`	bundle	$+0.012$	-0.012	n/a
15	all members self-trained (one teacher)	`final_B2`	`final_C`	bundle	-0.014	-0.049	n/a
16	more 1 s members	`final_B5`	`final_B6a`	bundle	-0.029	$+0.021$	n/a
17	wide CNN instead of 12-layer	`final_B5`	`final_F1`	bundle	-0.032	-0.001	$+$
18	sentence prior correction τ=0.3	`final_B5i`	`final_B5i_t0.3`	single	-0.037	$+0.017$	$+$
19	word-timing evidence β=0.5	`final_B5i`	`final_B5d05`	single	-0.060	-0.010	$+$
20	word-timing evidence β=1.0	`final_B5i`	`final_B5d10`	single	-0.024	-0.008	$+$
21	input rescaling	`final_B5i`	`final_B5irs`	single	-0.007	$+0.004$	n/a
22	validation-motivated stack V	`final_B5i`	`final_V`	bundle	-0.027	$+0.022$	$+$
23	lr-fixed deep members	`final_B5i`	`final_B8b`	bundle	-0.011	$+0.002$	n/a
24	in-domain LM (807 sentences, seed 0)	`final_B5i`	`final_B5i_L1`	single	$+0.010$	-0.038	n/a
25	in-domain LM, 3-seed mixture	`final_B5i`	`final_B5i_L1avg`	bundle	$+0.010$	-0.012	n/a
26	in-domain corpus 807 $\rightarrow$ 1,368	`final_B5i_L1`	`final_B5i_L3`	single	-0.010	$+0.038$	n/a
27	in-domain corpus 807 $\rightarrow$ 2,103 (adds Qwen; IN1)	`final_B5i_L1_IN1`	`final_IN1_Q_llm_plus`	single	$+0.002$	$+0.018$	n/a
28	in-domain corpus 2,103 $\rightarrow$ 3,742 (IN1)	`final_IN1_Q_llm_plus`	`final_IN1_Q_big`	single	$+0.004$	$+0.007$	n/a
29	generic candidates, λ=0.15	`final_IN1_Q_llm_plus`	`final_IN1Qp_SC0.15`	single	-0.003	$+0.012$	n/a
30	candidate decoding (guided pool + LM prior)	`final_IN1_Q_llm_plus`	`final_IN1Qp_SCGL_b0.5`	bundle	$+0.009$	-0.006	n/a